Image Classification using Fashion-MNIST image data (PyTorch)

```
# PyTorch
import torch
print(f"Is CUDA available? {torch.cuda.is_available()}")
print(f"GPU Name: {torch.cuda.get_device_name(0)}")
print(f"CUDA Version: {torch.version.cuda}")
import torchvision
import torchvision.transforms as transforms
import torch.nn as nn
import torch.nn.functional as nnFunc
import torch.amp as amp
# General
import numpy as np
import matplotlib.pyplot as plt
# Optimizer
import torch.optim as optim
# pretty print
from pprint import pprint
```

```
Is CUDA available? True
GPU Name: NVIDIA GeForce RTX 5050 Laptop GPU
CUDA Version: 13.0
```

Load image data

Fashion-MNIST is a dataset of Zalando's article images consisting of a training set of 60,000 examples and a test set of 10,000 examples. Each example is a 28x28 grayscale image, associated with a label from 10 classes.

https://github.com/zalandoresearch/fashion-mnist

1. Transform setting

The original images are in PIL (0 to 255 for each pixel). First we convert it to a tensor format (0 to 1). Then to make the data to be zero centering, we make the range to be between -1 to 1. If we use the following equation for each data, we can make it zero centered: \begin{equation} output = \frac{input - mean}{std} \end{equation} This is being done by setting the `mean` and `standard deviation` to be 0.5.

```
transform = transforms.Compose( [ transforms.RandomHorizontalFlip(),
                                  transforms.RandomCrop(28, padding=4),
                                  transforms.ToTensor(),
                                  transforms.Normalize((0.5,),
```

```
                                                                    (0.5,))
                                        ])
```

2. Making the training set

Now we load the actual dataset and make the training dataset.

1. Load MNIST dataset (Download and prepare dataset)
    a. Put it in `./data_FashionMNIST` directory.
    b. Download it
    c. Do the specified transformation on the dataset
2. Create the data loader (pipelining settings)
    a. Brake it to batches with size of 32
    b. Shuffle the deck in every epoch
    c. Use 4 number of workers to make the training faster

```
batchSize = 512
numberOfWorkers = 0

trainSet =
torchvision.datasets.FashionMNIST(root='./data_FashionMNIST',
                                                train=True,
                                                download=True,
                                                transform=transform)
trainLoader = torch.utils.data.DataLoader(trainSet,
                                            batch_size=batchSize,
                                            shuffle=True,
                                            num_workers=numberOfWorkers,
                                            pin_memory=True,
                                            persistent_workers=False)
```

3. Making the test set

Now making the test set. The procedure is the same as making the training set with some minor modifications.

```
testSet =
torchvision.datasets.FashionMNIST(root='./data_FashionMNIST',
                                          train=False, # Diff
                                          download=True,
                                          transform=transform)
testLoader = torch.utils.data.DataLoader(testSet,
                                            batch_size=batchSize,
                                            shuffle=False, # Diff
                                            num_workers=numberOfWorkers,
                                            pin_memory=True,
                                            persistent_workers=False)
```

```
viz_batchSize = 4
vizLoader = torch.utils.data.DataLoader(testSet,
                                        batch_size=viz_batchSize,
                                        shuffle=True)
```

4. The classes

```
classes = ['T-shirt/top', 'Trouser', 'Pullover', 'Dress', 'Coat',
'Sandal', 'Shirt', 'Sneaker', 'Bag', 'Ankle boot']
print(classes)
```

```
['T-shirt/top', 'Trouser', 'Pullover', 'Dress', 'Coat', 'Sandal',
'Shirt', 'Sneaker', 'Bag', 'Ankle boot']
```

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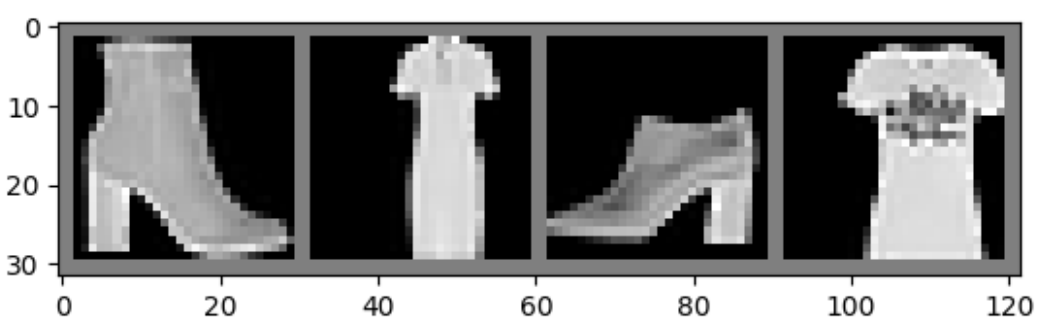
Showing some of the images from the dataset.

```
# functions to show an image
def imshow(img):
  img = img / 2 + 0.5 # unnormalize
  npimg = img.numpy()
  plt.imshow(np.transpose(npimg, (1, 2, 0)))
  plt.show()
# get some random training images
dataiter = iter(vizLoader)
images, labels = next(dataiter)
# show images
imshow(torchvision.utils.make_grid(images))
# print labels
pprint(' '.join(f'{classes[labels[j]]:5s}' for j in
range(viz_batchSize)))
```




```
'Ankle boot Dress Ankle boot T-shirt/top'
```

5. Defining the CNN model

```
class Net(nn.Module):
  def __init__(self):
    super().__init__()
    self.conv1    = nn.Conv2d(1, 6, 5, padding='same')
    self.pool     = nn.MaxPool2d(2, 2)
    self.conv2    = nn.Conv2d(6, 16, 5)
    self.fc1      = nn.Linear(16 * 5 * 5, 128)
    self.fc2      = nn.Linear(128, 64)
    self.dropout  = nn.Dropout(0.2)
    self.fc3      = nn.Linear(64, 10)

  def forward(self, x):
    x = self.pool(nnFunc.relu(self.conv1(x)))
    x = self.pool(nnFunc.relu(self.conv2(x)))
    x = x.view(-1, 16 * 5 * 5)
    x = nnFunc.relu(self.fc1(x))
    x = nnFunc.relu(self.fc2(x))
    x = self.dropout(x)
    x = self.fc3(x)
    return x
```

6. Instansiating the CNN model

```
def init_weights(m):
    if isinstance(m, nn.Conv2d) or isinstance(m, nn.Linear):
        nn.init.kaiming_normal_(m.weight, mode='fan_out',
nonlinearity='relu')
        if m.bias is not None:
            nn.init.constant_(m.bias, 0)
net = Net()
net.apply(init_weights)
```

```
Net(
  (conv1): Conv2d(1, 6, kernel_size=(5, 5), stride=(1, 1),
padding=same)
  (pool): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
  (conv2): Conv2d(6, 16, kernel_size=(5, 5), stride=(1, 1))
  (fc1): Linear(in_features=400, out_features=128, bias=True)
  (fc2): Linear(in_features=128, out_features=64, bias=True)
  (dropout): Dropout(p=0.2, inplace=False)
  (fc3): Linear(in_features=64, out_features=10, bias=True)
)
```

7. Move the model to GPU

```
device = torch.device('cuda:0' if torch.cuda.is_available() else
'cpu')
print(device)
```

```
cuda:0
```

```
net.to(device)
```

```
Net(
  (conv1): Conv2d(1, 6, kernel_size=(5, 5), stride=(1, 1),
padding=same)
  (pool): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
  (conv2): Conv2d(6, 16, kernel_size=(5, 5), stride=(1, 1))
  (fc1): Linear(in_features=400, out_features=128, bias=True)
  (fc2): Linear(in_features=128, out_features=64, bias=True)
  (dropout): Dropout(p=0.2, inplace=False)
  (fc3): Linear(in_features=64, out_features=10, bias=True)
)
```

7. Degine loss function and optimizer

```
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(net.parameters(), lr=0.001)
scheduler = optim.lr_scheduler.ReduceLROnPlateau(optimizer,
mode='max', factor=0.1, patience=10)
```

8. Traing the network

```
scaler = amp.GradScaler('cuda')

epochs = 30
printFrequency = len(trainLoader) // 4

for epoch in range(epochs):
    net.train()
    running_loss = 0.0
    correct = 0
    total = 0

    # --- TRAINING PHASE ---
    for i, data in enumerate(trainLoader, 0):
        inputs, labels = data[0].to(device, non_blocking=True),
data[1].to(device, non_blocking=True)

        optimizer.zero_grad()

        with amp.autocast('cuda'):
```

```
            outputs = net(inputs)
            loss = criterion(outputs, labels)

        scaler.scale(loss).backward()
        scaler.unscale_(optimizer)
        torch.nn.utils.clip_grad_norm_(net.parameters(), max_norm=1.0)

        scaler.step(optimizer)
        scaler.update()

        _, predicted = torch.max(outputs.data, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()
        running_loss += loss.item()

        if i % printFrequency == (printFrequency - 1):
            accuracy = 100 * correct / total
            print(f'Epoch {epoch + 1} | Batch {i +
1}/{len(trainLoader)} | loss: {running_loss / printFrequency:.3f} |
acc: {accuracy:.2f}%')
            running_loss, correct, total = 0.0, 0, 0

    # --- VALIDATION PHASE ---
    net.eval()
    val_correct = 0
    val_total = 0
    with torch.no_grad():
        for data in testLoader:
            images, labels = data[0].to(device), data[1].to(device)
            with amp.autocast('cuda'):
                outputs = net(images)
            _, predicted = torch.max(outputs.data, 1)
            val_total += labels.size(0)
            val_correct += (predicted == labels).sum().item()

    val_acc = 100 * val_correct / val_total
    current_lr = optimizer.param_groups[0]['lr']
    print(f'>> Epoch {epoch + 1} SUMMARY: Val Acc: {val_acc:.2f}% |
Current LR: {current_lr}')

    # --- AUTOMATIC LR DROP ---
    scheduler.step(val_acc)

print('Finished Training')
```

```
Epoch 1 | Batch 29/118 | loss: 2.705 | acc: 24.39%
Epoch 1 | Batch 58/118 | loss: 1.391 | acc: 50.09%
Epoch 1 | Batch 87/118 | loss: 1.111 | acc: 60.70%
Epoch 1 | Batch 116/118 | loss: 0.965 | acc: 65.45%
>> Epoch 1 SUMMARY: Val Acc: 71.27% | Current LR: 0.001
```

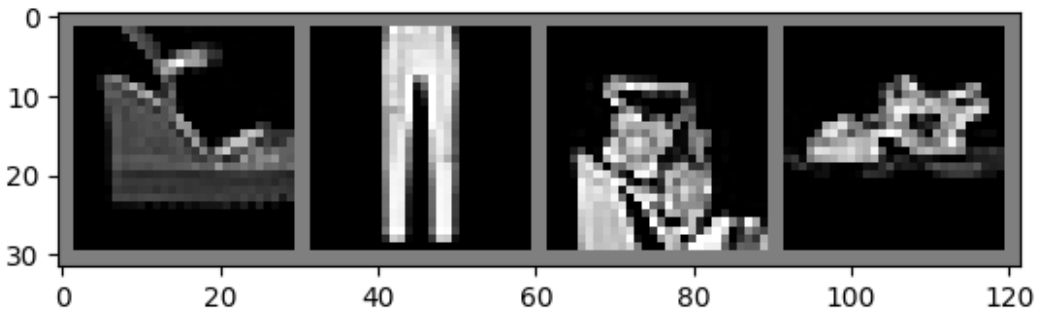
```
Epoch 2 | Batch 29/118 | loss: 0.904 | acc: 67.08%
Epoch 2 | Batch 58/118 | loss: 0.845 | acc: 69.69%
Epoch 2 | Batch 87/118 | loss: 0.807 | acc: 70.50%
Epoch 2 | Batch 116/118 | loss: 0.782 | acc: 71.43%
>> Epoch 2 SUMMARY: Val Acc: 75.02% | Current LR: 0.001
Epoch 3 | Batch 29/118 | loss: 0.746 | acc: 72.53%
Epoch 3 | Batch 58/118 | loss: 0.725 | acc: 73.25%
Epoch 3 | Batch 87/118 | loss: 0.712 | acc: 73.57%
Epoch 3 | Batch 116/118 | loss: 0.704 | acc: 74.04%
>> Epoch 3 SUMMARY: Val Acc: 76.59% | Current LR: 0.001
Epoch 4 | Batch 29/118 | loss: 0.690 | acc: 74.22%
Epoch 4 | Batch 58/118 | loss: 0.670 | acc: 75.22%
Epoch 4 | Batch 87/118 | loss: 0.656 | acc: 75.73%
Epoch 4 | Batch 116/118 | loss: 0.651 | acc: 75.90%
>> Epoch 4 SUMMARY: Val Acc: 77.18% | Current LR: 0.001
Epoch 5 | Batch 29/118 | loss: 0.639 | acc: 76.19%
Epoch 5 | Batch 58/118 | loss: 0.620 | acc: 76.63%
Epoch 5 | Batch 87/118 | loss: 0.623 | acc: 76.42%
Epoch 5 | Batch 116/118 | loss: 0.624 | acc: 76.88%
>> Epoch 5 SUMMARY: Val Acc: 78.50% | Current LR: 0.001
Epoch 6 | Batch 29/118 | loss: 0.611 | acc: 77.27%
Epoch 6 | Batch 58/118 | loss: 0.601 | acc: 77.52%
Epoch 6 | Batch 87/118 | loss: 0.603 | acc: 77.38%
Epoch 6 | Batch 116/118 | loss: 0.591 | acc: 77.85%
>> Epoch 6 SUMMARY: Val Acc: 79.63% | Current LR: 0.001
Epoch 7 | Batch 29/118 | loss: 0.590 | acc: 77.60%
Epoch 7 | Batch 58/118 | loss: 0.575 | acc: 78.63%
Epoch 7 | Batch 87/118 | loss: 0.567 | acc: 78.96%
Epoch 7 | Batch 116/118 | loss: 0.566 | acc: 79.22%
>> Epoch 7 SUMMARY: Val Acc: 79.79% | Current LR: 0.001
Epoch 8 | Batch 29/118 | loss: 0.547 | acc: 79.65%
Epoch 8 | Batch 58/118 | loss: 0.559 | acc: 78.67%
Epoch 8 | Batch 87/118 | loss: 0.549 | acc: 79.42%
Epoch 8 | Batch 116/118 | loss: 0.556 | acc: 79.17%
>> Epoch 8 SUMMARY: Val Acc: 80.81% | Current LR: 0.001
Epoch 9 | Batch 29/118 | loss: 0.539 | acc: 80.31%
Epoch 9 | Batch 58/118 | loss: 0.534 | acc: 79.70%
Epoch 9 | Batch 87/118 | loss: 0.540 | acc: 80.19%
Epoch 9 | Batch 116/118 | loss: 0.540 | acc: 79.91%
>> Epoch 9 SUMMARY: Val Acc: 81.40% | Current LR: 0.001
Epoch 10 | Batch 29/118 | loss: 0.545 | acc: 79.88%
Epoch 10 | Batch 58/118 | loss: 0.521 | acc: 80.99%
Epoch 10 | Batch 87/118 | loss: 0.526 | acc: 80.72%
Epoch 10 | Batch 116/118 | loss: 0.516 | acc: 80.66%
>> Epoch 10 SUMMARY: Val Acc: 82.30% | Current LR: 0.001
Epoch 11 | Batch 29/118 | loss: 0.515 | acc: 81.10%
Epoch 11 | Batch 58/118 | loss: 0.514 | acc: 80.90%
Epoch 11 | Batch 87/118 | loss: 0.513 | acc: 80.60%
Epoch 11 | Batch 116/118 | loss: 0.508 | acc: 81.53%
```

```
>> Epoch 11 SUMMARY: Val Acc: 80.82% | Current LR: 0.001
Epoch 12 | Batch 29/118 | loss: 0.512 | acc: 80.95%
Epoch 12 | Batch 58/118 | loss: 0.508 | acc: 81.29%
Epoch 12 | Batch 87/118 | loss: 0.495 | acc: 81.20%
Epoch 12 | Batch 116/118 | loss: 0.495 | acc: 81.62%
>> Epoch 12 SUMMARY: Val Acc: 82.92% | Current LR: 0.001
Epoch 13 | Batch 29/118 | loss: 0.485 | acc: 82.15%
Epoch 13 | Batch 58/118 | loss: 0.486 | acc: 81.83%
Epoch 13 | Batch 87/118 | loss: 0.510 | acc: 80.99%
Epoch 13 | Batch 116/118 | loss: 0.489 | acc: 81.88%
>> Epoch 13 SUMMARY: Val Acc: 83.19% | Current LR: 0.001
Epoch 14 | Batch 29/118 | loss: 0.488 | acc: 81.90%
Epoch 14 | Batch 58/118 | loss: 0.473 | acc: 82.28%
Epoch 14 | Batch 87/118 | loss: 0.482 | acc: 82.02%
Epoch 14 | Batch 116/118 | loss: 0.485 | acc: 82.11%
>> Epoch 14 SUMMARY: Val Acc: 82.97% | Current LR: 0.001
Epoch 15 | Batch 29/118 | loss: 0.477 | acc: 82.31%
Epoch 15 | Batch 58/118 | loss: 0.468 | acc: 82.66%
Epoch 15 | Batch 87/118 | loss: 0.482 | acc: 82.46%
Epoch 15 | Batch 116/118 | loss: 0.469 | acc: 82.64%
>> Epoch 15 SUMMARY: Val Acc: 82.98% | Current LR: 0.001
Epoch 16 | Batch 29/118 | loss: 0.463 | acc: 82.94%
Epoch 16 | Batch 58/118 | loss: 0.467 | acc: 82.29%
Epoch 16 | Batch 87/118 | loss: 0.458 | acc: 83.19%
Epoch 16 | Batch 116/118 | loss: 0.465 | acc: 83.12%
>> Epoch 16 SUMMARY: Val Acc: 82.91% | Current LR: 0.001
Epoch 17 | Batch 29/118 | loss: 0.474 | acc: 82.36%
Epoch 17 | Batch 58/118 | loss: 0.453 | acc: 83.16%
Epoch 17 | Batch 87/118 | loss: 0.451 | acc: 83.39%
Epoch 17 | Batch 116/118 | loss: 0.455 | acc: 82.79%
>> Epoch 17 SUMMARY: Val Acc: 83.55% | Current LR: 0.001
Epoch 18 | Batch 29/118 | loss: 0.467 | acc: 82.86%
Epoch 18 | Batch 58/118 | loss: 0.447 | acc: 83.30%
Epoch 18 | Batch 87/118 | loss: 0.444 | acc: 83.38%
Epoch 18 | Batch 116/118 | loss: 0.444 | acc: 83.84%
>> Epoch 18 SUMMARY: Val Acc: 84.00% | Current LR: 0.001
Epoch 19 | Batch 29/118 | loss: 0.443 | acc: 83.82%
Epoch 19 | Batch 58/118 | loss: 0.443 | acc: 83.37%
Epoch 19 | Batch 87/118 | loss: 0.458 | acc: 83.28%
Epoch 19 | Batch 116/118 | loss: 0.447 | acc: 83.36%
>> Epoch 19 SUMMARY: Val Acc: 83.61% | Current LR: 0.001
Epoch 20 | Batch 29/118 | loss: 0.438 | acc: 83.94%
Epoch 20 | Batch 58/118 | loss: 0.447 | acc: 83.54%
Epoch 20 | Batch 87/118 | loss: 0.444 | acc: 83.38%
Epoch 20 | Batch 116/118 | loss: 0.429 | acc: 84.17%
>> Epoch 20 SUMMARY: Val Acc: 84.37% | Current LR: 0.001
Epoch 21 | Batch 29/118 | loss: 0.434 | acc: 84.11%
Epoch 21 | Batch 58/118 | loss: 0.441 | acc: 83.59%
Epoch 21 | Batch 87/118 | loss: 0.429 | acc: 84.27%
```

```
Epoch 21 | Batch 116/118 | loss: 0.431 | acc: 83.95%
>> Epoch 21 SUMMARY: Val Acc: 84.52% | Current LR: 0.001
Epoch 22 | Batch 29/118 | loss: 0.439 | acc: 83.78%
Epoch 22 | Batch 58/118 | loss: 0.426 | acc: 84.38%
Epoch 22 | Batch 87/118 | loss: 0.431 | acc: 83.97%
Epoch 22 | Batch 116/118 | loss: 0.423 | acc: 84.15%
>> Epoch 22 SUMMARY: Val Acc: 84.50% | Current LR: 0.001
Epoch 23 | Batch 29/118 | loss: 0.421 | acc: 84.66%
Epoch 23 | Batch 58/118 | loss: 0.440 | acc: 83.80%
Epoch 23 | Batch 87/118 | loss: 0.422 | acc: 84.26%
Epoch 23 | Batch 116/118 | loss: 0.425 | acc: 84.33%
>> Epoch 23 SUMMARY: Val Acc: 84.12% | Current LR: 0.001
Epoch 24 | Batch 29/118 | loss: 0.419 | acc: 84.68%
Epoch 24 | Batch 58/118 | loss: 0.433 | acc: 83.80%
Epoch 24 | Batch 87/118 | loss: 0.421 | acc: 84.43%
Epoch 24 | Batch 116/118 | loss: 0.405 | acc: 84.87%
>> Epoch 24 SUMMARY: Val Acc: 84.55% | Current LR: 0.001
Epoch 25 | Batch 29/118 | loss: 0.416 | acc: 84.44%
Epoch 25 | Batch 58/118 | loss: 0.414 | acc: 84.64%
Epoch 25 | Batch 87/118 | loss: 0.413 | acc: 84.58%
Epoch 25 | Batch 116/118 | loss: 0.417 | acc: 84.57%
>> Epoch 25 SUMMARY: Val Acc: 84.29% | Current LR: 0.001
Epoch 26 | Batch 29/118 | loss: 0.415 | acc: 84.65%
Epoch 26 | Batch 58/118 | loss: 0.408 | acc: 84.97%
Epoch 26 | Batch 87/118 | loss: 0.424 | acc: 84.56%
Epoch 26 | Batch 116/118 | loss: 0.401 | acc: 84.99%
>> Epoch 26 SUMMARY: Val Acc: 85.16% | Current LR: 0.001
Epoch 27 | Batch 29/118 | loss: 0.400 | acc: 85.30%
Epoch 27 | Batch 58/118 | loss: 0.410 | acc: 84.98%
Epoch 27 | Batch 87/118 | loss: 0.420 | acc: 84.52%
Epoch 27 | Batch 116/118 | loss: 0.413 | acc: 85.24%
>> Epoch 27 SUMMARY: Val Acc: 85.37% | Current LR: 0.001
Epoch 28 | Batch 29/118 | loss: 0.397 | acc: 85.49%
Epoch 28 | Batch 58/118 | loss: 0.404 | acc: 85.14%
Epoch 28 | Batch 87/118 | loss: 0.409 | acc: 85.11%
Epoch 28 | Batch 116/118 | loss: 0.403 | acc: 85.12%
>> Epoch 28 SUMMARY: Val Acc: 85.14% | Current LR: 0.001
Epoch 29 | Batch 29/118 | loss: 0.405 | acc: 84.91%
Epoch 29 | Batch 58/118 | loss: 0.406 | acc: 84.85%
Epoch 29 | Batch 87/118 | loss: 0.401 | acc: 85.34%
Epoch 29 | Batch 116/118 | loss: 0.399 | acc: 85.26%
>> Epoch 29 SUMMARY: Val Acc: 85.86% | Current LR: 0.001
Epoch 30 | Batch 29/118 | loss: 0.403 | acc: 84.99%
Epoch 30 | Batch 58/118 | loss: 0.403 | acc: 84.85%
Epoch 30 | Batch 87/118 | loss: 0.398 | acc: 85.49%
Epoch 30 | Batch 116/118 | loss: 0.390 | acc: 85.10%
>> Epoch 30 SUMMARY: Val Acc: 85.85% | Current LR: 0.001
Finished Training
```

9. Test the network on the test data

```
dataIter = iter(vizLoader)
images, labels = next(dataIter)
# print images
imshow(torchvision.utils.make_grid(images))
print('GroundTruth: ', ' '.join(f'{classes[labels[j]]:5s}' for j in
range(viz_batchSize)))
```




```
GroundTruth:  Sandal Trouser Ankle boot Sandal

# Move the data to GPU
images, labels = images.to(device), labels.to(device)

# Now you can run it
outputs = net(images)
_, predicted = torch.max(outputs, 1)

# IMPORTANT: To print, move back to CPU
# Printing needs the labels to be on the CPU so Python can read them
print('Predicted: ', ' '.join(f'{classes[predicted[j].item()]:5s}' for
j in range(len(predicted))))

Predicted:  Sandal Trouser Sandal Sandal
```

10. Test the network on the whole dataset

```
test_loss = 0.0
correct = 0
total = 0

with torch.no_grad():
    for data in testLoader:
        images, labels = data[0].to(device), data[1].to(device)

        outputs = net(images)

        # 1. Calculate the loss for this batch
        loss = criterion(outputs, labels)
```

```
        test_loss += loss.item()

        # 2. Calculate accuracy as before
        _, predicted = torch.max(outputs, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()

# Calculate the average loss over all batches
avg_test_loss = test_loss / len(testLoader)

print(f'Test Loss: {avg_test_loss:.4f}')
print(f'Test Accuracy: {100 * correct / total:.2f}%')
```

```
Test Loss: 0.3830
Test Accuracy: 85.93%
```

11. Check performance within each class

```
# prepare to count predictions for each class
correct_pred = {classname: 0 for classname in classes}
total_pred = {classname: 0 for classname in classes}

# again no gradients needed
with torch.no_grad():
    for data in testLoader:
        images, labels = data[0].to(device), data[1].to(device)
        outputs = net(images)
        _, predictions = torch.max(outputs, 1)
        # collect the correct predictions for each class
        for label, prediction in zip(labels, predictions):
            if label == prediction:
                correct_pred[classes[label]] += 1
            total_pred[classes[label]] += 1

# print accuracy for each class
for classname, correct_count in correct_pred.items():
    accuracy = 100 * float(correct_count) / total_pred[classname]
    print(f'Accuracy for class: {classname:5s} is {accuracy:.1f} %')
```

```
Accuracy for class: T-shirt/top is 87.1 %
Accuracy for class: Trouser is 96.7 %
Accuracy for class: Pullover is 81.4 %
Accuracy for class: Dress is 88.3 %
Accuracy for class: Coat  is 73.2 %
Accuracy for class: Sandal is 90.9 %
Accuracy for class: Shirt is 51.5 %
Accuracy for class: Sneaker is 93.8 %
Accuracy for class: Bag   is 95.3 %
Accuracy for class: Ankle boot is 95.4 %
```